Lecture 26

Last time : * Graphical analysis for energy conversion

* $EFE = i\,d\lambda$

$EFM = -\int \tau_e\, d\theta$ or $\int -f_e\, dx$.

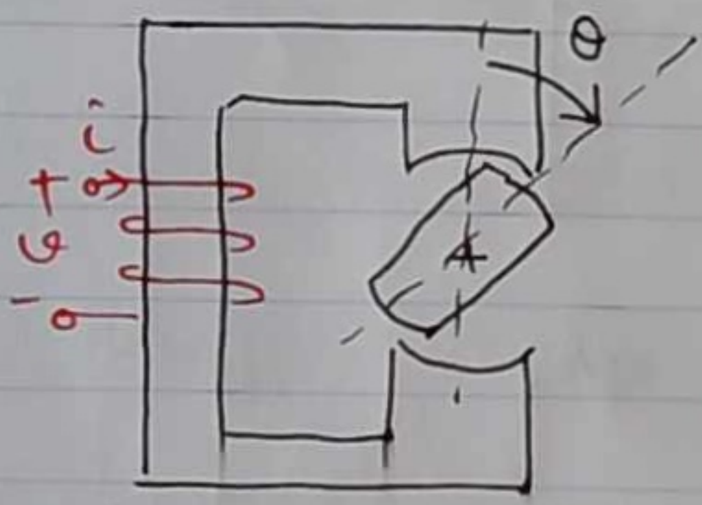
* $dW_m = EFE + EFM$.

So, if we move from point A to B.

$$\int_A^B dW_m = \int_A^B i\,d\lambda + \int_A^B -\tau_e\, d\theta$$

$$W_m|_B - W_m|_A = \underset{A\to B}{EFE} + \underset{A\to B}{EFM}$$

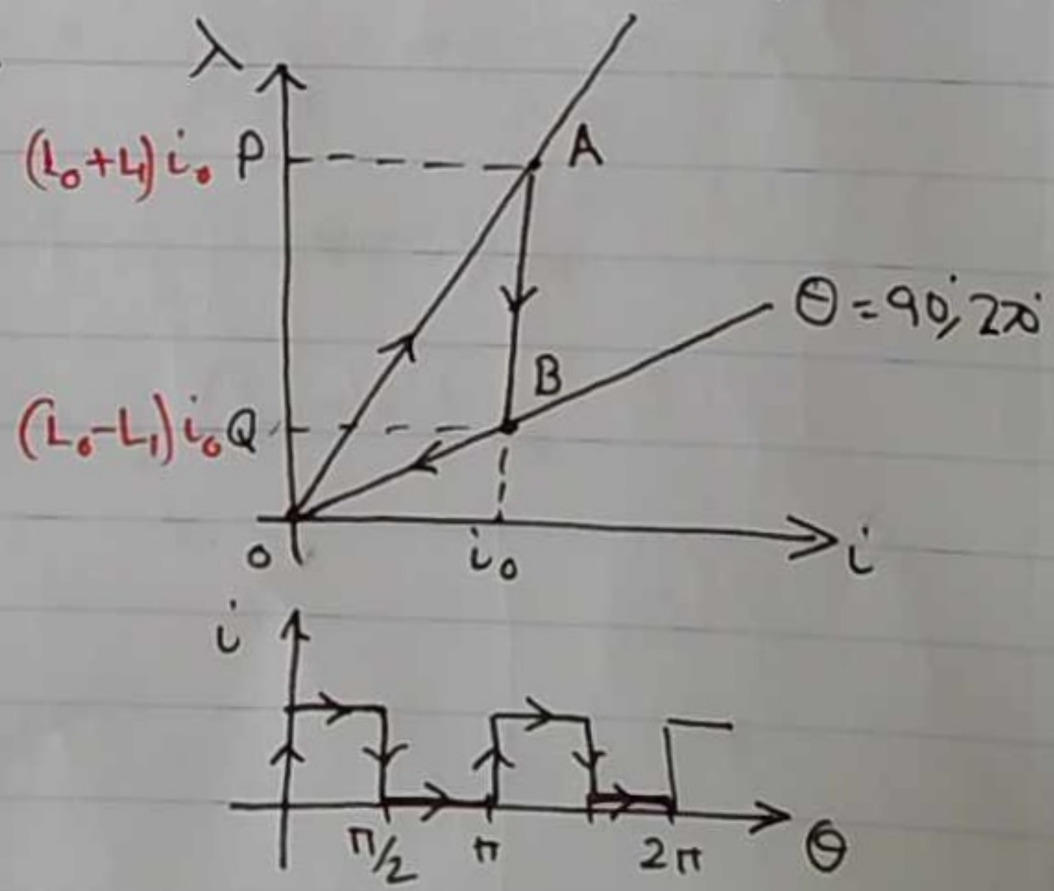
Today : We will run our reluctance machine as generator. How? By running the cycle reverse.



$L(\theta) = L_0 + L_1 \cos 2\theta$.

$\lambda = L(\theta)\, i$

$\tau_e = -i^2 L_1 \sin 2\theta$



OA

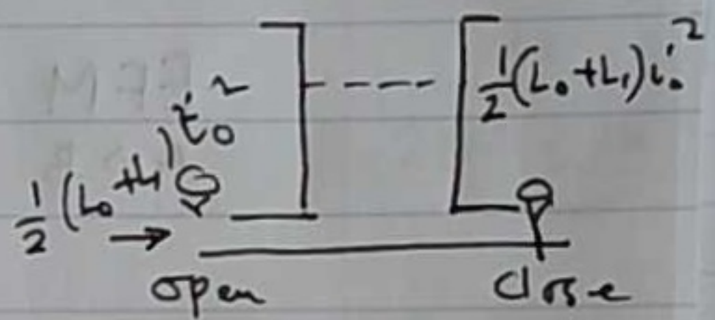
$$d\omega_m = i\,d\lambda - \tau_e\,d\theta$$

$\uparrow$ zero as θ = const

$$\therefore \quad \omega_m\Big|_A = \int i\,d\lambda = \frac{\lambda_A^2}{2L(0°)} = \frac{1}{2}(L_0+L_1)\,i_0^2$$

$$\underset{O\to A}{EFE} = \frac{1}{2}(L_0+L_1)\,i_0^2$$

$$= \Delta OAP$$



AB

$$\int_A^B d\omega_m = \int_A^B i\,d\lambda - \int_A^B \tau_e\,d\theta$$

$$\Rightarrow \omega_m\Big|_B - \omega_m\Big|_A = \underset{A\to B}{EFE} + \underset{A\to B}{EFM}$$

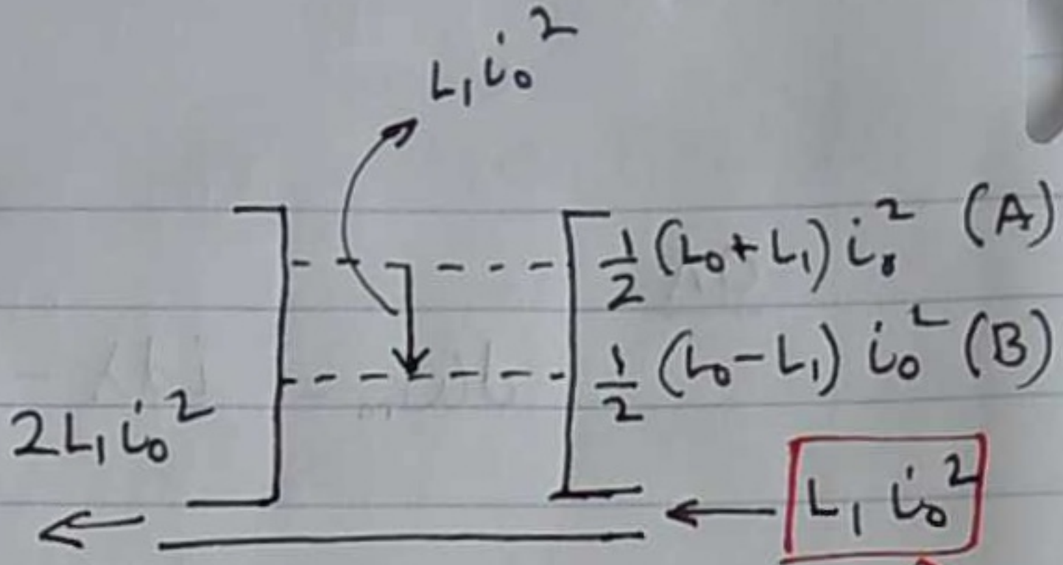
(a) $\omega_m\Big|_B = \Delta OBQ = \frac{1}{2}(L_0+L_1)\,i_0^2$

$\omega_m\Big|_A = \Delta OAP = \frac{1}{2}(L_0+L_1)\,i_0^2$

$$\therefore \quad \underset{A\to B}{\Delta\omega_m} = -L_1 i_0^2$$

$$\underset{A\to B}{EFE} = \int_A^B i\,d\lambda = \square ABQP = -2L_1 i_0^2$$

$$\underset{A\to B}{EFM} = L_1 i_0^2$$



Here, EFM is calculated indirectly, can we directly calculate EFM?

$$\underset{A \to B}{EFM} = -\int_A^B \tau_e \, d\theta = -i_0^2 L_1 \int_{0^\circ}^{90^\circ} \sin 2\theta \, d\theta$$

Matches

$$= \frac{L_1 i_0^2}{2} \left[\cos 2\theta\right]_{0^\circ}^{90^\circ}$$

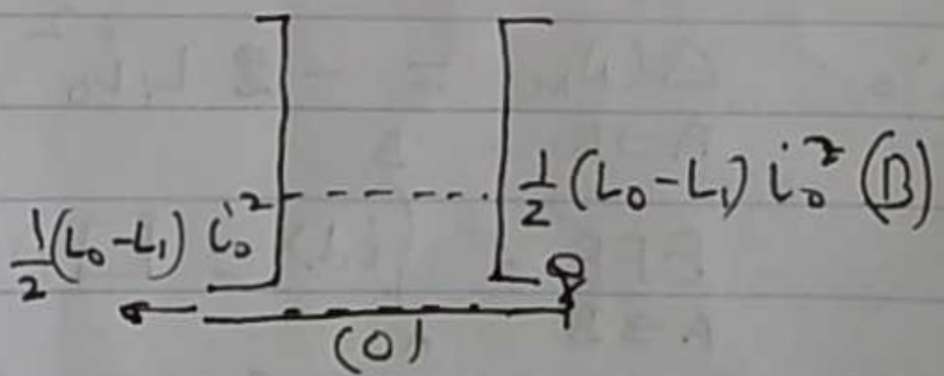
$$= \boxed{-L_1 i_0^2}$$

B → O

$EFM = 0$; const θ

$$\int_B^0 d\omega_m = \int_B^0 i \, d\lambda + 0$$

$$\therefore \quad \omega_m|_0 - \omega_B|_B = \frac{B}{2} \Delta OBQ = \frac{-1}{2}(L_0 - L_1) i_0^2$$

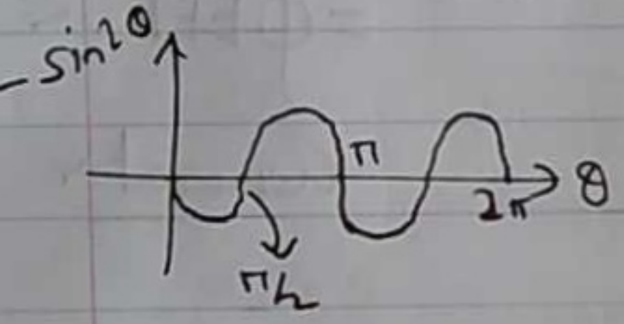


Q. How will the voltage profile look like at the electrical port ?

$$v = \frac{d\lambda}{dt} = \frac{d}{dt}(L(\theta) i) = L(\theta)\frac{di}{dt} + i\frac{\partial L(\theta)}{\partial \theta}\frac{d\theta}{dt}.$$

OA : $\theta = \text{const} \therefore \frac{d\theta}{dt} = 0 \therefore v = (L_0 + L_1)\frac{di}{dt}$.

AB : $i = \text{const} \therefore \frac{di}{dt} = 0 \therefore v = -2L_1 \sin 2\theta \cdot i_0 \frac{d\theta}{dt}$.



$= -2L_1 \sin 2\theta \cdot i_0 \frac{d\theta}{dt}$.

BO : $\theta = \text{const} \therefore \frac{d\theta}{dt} = 0$ and $v = (L_0 - L_1)\frac{di}{dt}$.

